International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Semester Final Examination, Spring-2025

Course Code: CSE 3521 Course Title: Computer Architecture

Total marks: **50** Time: **2.5 hours**

<table>
<tr><th></th><th>Group-A</th><th>M</th><th>CO</th></tr>
<tr><td>1. a)</td><td>Suppose you are given an instruction add $s1,$t3,$t4. Describe the complete operation of the datapath using the instruction with a figure.</td><td>5</td><td>3</td></tr>
<tr><td>b)</td><td>Design a multi-cycle data path for handling basic instruction of MIPS.</td><td>5</td><td>3</td></tr>
<tr><td>2. a)</td><td>How does the execution of different instruction types differ in multi-cycle datapath. Show the steps.</td><td>5</td><td>3</td></tr>
<tr><td>b)</td><td>An instruction set consists of the following instruction mix:
35% ALU instructions; 20% Load instructions; 15% Store instructions; 25% Branch instructions; 5% Jump instructions
1. Calculate the overall CPI for this instruction mix in a variable-cycle implementation.
2. If the processor is redesigned with a fixed-cycle, where every instruction takes 3 cycles, what is the new CPI?
3. Compare the CPI in both cases and discuss the trade-offs between using a variable-length cycle design versus a fixed-length cycle design.</td><td>5</td><td>2</td></tr>
<tr><td></td><td>OR</td><td></td><td></td></tr>
<tr><td>2. a)</td><td>What changes are required for the multiplexer in multicycle Datapath?</td><td>5</td><td>3</td></tr>
<tr><td>b)</td><td>A multicycle CPU has a clock time of 250 ps and the following instruction mix:
<table><tr><th>Instruction</th><th>%</th><th>Cycles</th></tr><tr><td>R-type</td><td>60</td><td>4</td></tr><tr><td>lw</td><td>40</td><td>5</td></tr></table>The program has 100,000 instructions.
Q: What is the total CPU time?</td><td>5</td><td>2</td></tr>
<tr><td></td><td>Group-B</td><td></td><td></td></tr>
<tr><td>3.a)</td><td>Explain the concept of pipeline stalls caused by data hazards. Propose two techniques to resolve or minimize data hazards.</td><td>5</td><td>3</td></tr>
<tr><td>b)</td><td>Draw and explain the pipeline stages involved in executing a store instruction in a pipelined architecture. Discuss how hazards are handled during execution.</td><td>5</td><td>3</td></tr>
<tr><td></td><td>OR</td><td></td><td></td></tr>
<tr><td>3.a)</td><td>Compare and contrast pipelined and single-cycle instruction execution.Use timing diagrams and MIPS instruction examples to support your explanation. Include assumptions about stage delays and overall speedup.</td><td>5</td><td>3</td></tr>
<tr><td>b)</td><td>Design a pipeline schedule for the following MIPS instructions to minimize hazards:
lw $t1, 0($t2)
add $t3, $t1, $t4
sub $t5, $t3, $t6
a) Identify the types of data hazards present.
b) Suggest how forwarding or stalling could resolve them.
c) Reorder the instructions (if possible) to minimize stalls.</td><td>5</td><td>3</td></tr>
<tr><td>4.a)</td><td>Discuss the working of a direct-mapped cache. Include the addressing scheme and explain how hits and misses are determined.</td><td>5</td><td>2</td></tr>
<tr><td>b)</td><td>What is virtual memory? How does the page table help translate virtual addresses into physical addresses?</td><td>5</td><td>2</td></tr>
<tr><td>5. a)</td><td>Describe how Translation Lookaside Buffer (TLB) improves memory access time. What happens on a TLB miss?</td><td>5</td><td>2</td></tr>
<tr><td>b)</td><td>What is handshaking in asynchronous communication? Describe its steps with an example of a read request from an I/O device.</td><td>5</td><td>2</td></tr>
</table>

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International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Autumn 2024

Course Code: CSE 3521 Course Title: Computer Architecture

Total marks: **50** Time: **2.5 hours**

[Answer all the questions. Figures in the right hand margin indicate full marks. M:Marks, C:CLO, D: Domain]

		M	C	D
	Group-A			
1 a)	Compare the performance of single-cycle, multi-cycle, and pipeline processors in terms of CPI. Discuss with a detailed example.	5	5	1
	OR			
	Suppose you are given an instruction add $s1,$t3,$t4. Describe the complete operation of the datapath using the instruction with a figure.			
b)	Explain the structural differences between single and multi-cycle data path.	5	5	2
	OR			
	What temporary registers are used in multicycle datapath? What changes are required for the multiplexer in multicycle datapath?			
2. a)	Differentiate the single and multi-cycle data path on the basis of their control signal.	5	5	2
b)	A program has the following instruction mix: 30% ALU operations, 25% loads, 10% stores, 30% branches, and 5% jumps. If each instruction type has the following cycle counts: ALU - 1, Load - 2, Store - 2, Branch - 1, Jump - 1, calculate the overall CPI.	5	2	3
	Group-B			
3.a)	What is the role of control hazards in pipeline architecture? Propose a method to minimize control hazards.	5	5	2
b)	Design a diagram showing the working of a branch instruction in a pipelined data path and explain the steps involved.	5	5	2
4.a)	Define *miss penalty, cache miss, hit rate, tag* in caching, memory hierarchy	5	3	2
b)	What is DMA controller? Describe the configuration of DMA controller.	5	3	3
5. a)	What is TLB? Discuss the operation of paging and TLB using flow diagram?	5	3	2
b)	Explain the direct mapped cache system with figure.	5	4	3
	OR			
a)	Explain the role of virtual memory in modern computer systems. Describe the translation process of virtual addresses to physical addresses with the help of a diagram.	5	3	2
b)	What is RAID? Explain at least three of its type with example.	5	4	3

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Total marks: **50** Time: **2.5 hours**

Answer all Five of the following questions. The figures in the right-hand margin indicate full marks.

Group - A

		M	C	D
1.a)	Describe the edge triggered clocking methodology.	4	1	
b)	Suppose you are given an instruction add $s1,$t3,$t4. Describe the complete operation of the datapath using the instruction with a figure.	4	1	
c)	Differentiate multicycle over single cycle datapath?	2	2	
2.a)	Points out different steps and benefits of fetch execute cycle	3	5	
b)	Design and describe the working procedure of jump instruction in single-cycle data path.	7	5	
	OR			
2.a)	Write down the basic characteristics of RISc and CISC architectures .	4	1	
b)	What is exception? How exception are handled?	4	1	
c)	What temporary registers are used in multicycle datapath?	2	2	

Group - B

3 a)	Does the use of pipelining reduce, increase or leave unchanged the execution time of a single machine instruction?	3	3	
b)	How can you resolve data hazard -explain with figure.	5	1	
c)	Dynamic pipelining is more complicated than the traditional or static pipelining – why?	2	2	
4.a)	How does spatial locality be helpful in handling cache misses? Explain with example.	3	3	
b)	Explain the working principle of direct mapped cache with figure. How many total bits are needed for a direct mapped cache with 64KB data in 1 word blocks?	7	3	
5 a)	Explain the working principle of TLB with figures.	5	3	
b)	What is the difference between synchronous bus and asynchronous bus? We want to compare the maximum bandwidth for a sync and async bus. The sync bus has a clock cycle time of 50ns and each bus transmission takes 1 cycle. The async bus requires 40ns per handshake. Each bus has 32 data lines. What is the bandwidth when performing one word reads from a 200ns memory?	5	4	
	OR			
5 a)	Explain cache read hit/miss.	2	1	
b)	Define miss penalty, cache miss, hit rate, tag in caching.	4	1	
c)	What is DMA controller? Describe the configuration of DMA controller.	4	1	

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Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Semester Final Examination, Autumn 2023

Course Code:CSE 3521 Course Title: Computer Architecture

Total marks: 50 Time: 2 hours 30 minutes

[Answer all questions from **Group-A** and **Group-B**
Separate answer script must be used for Group-A and Group-B.]

Group-A

1.a)	What is hardwired control? How it is different from micro-programmed control?	2	CO1
b)	Suppose you are given an instruction **sub $s1,$t3,$t4**. Describe the complete operation of the datapath using the instruction with a figure.	5	CO1
	Or,		
	Illustrate the following instruction with three bus and two bus CPU system and explain which is better? ADD R3, R2, R1		
c)	Why a single-cycle implementation is not used? justify the answer.	3	CO2
2.a)	Do you think that the functions of memory data register and instruction register are the same? If not then how?	3	CO
b)	Write down the basic characteristics of RISC and CISC architectures.	5	CO
	Or,		
	Design a multi-cycle data path for handling basic instruction of MIPS.		
c)	What changes are required for the multiplexer in multicycle datapath?	2	CO

Group-B

3.a)	Does the use of pipelining reduce, increase or leave unchanged the execution time of a single machine instruction?	3	CO
b)	How can you resolve data hazard -explain with figure.	5	CO
c)	Dynamic pipelining is more complicated than the traditional or static pipelining – why?	2	CO2
4.a)	Remark on the purpose, scope and mechanisms of the followings: i)Programmed I/O ii)Interrupt Initiated I/O iii) Direct Memory Access.	3	CO4
b)	What is a DMA controller? Write the configuration of the DMA controller.	5	CO1
	Or		
	What is TLB? How can you reduce page fault? Explain logically?	2	CO1
c)	Define miss penalty, cache miss, hit rate, tag in caching.		

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5.a)	Find the Location of a memory block with address 14 in a cache with 8 blocks with different degrees of associativity	5	CO2
	Or		
	When will the page faults occur? What is the procedure for handling the page fault?		
b)	What are the key factors for designing a virtual memory? What are similarities between cache and virtual memory?	3	CO
c)	Show the comparative analysis of write through scheme and write back scheme.	2	CO

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Total marks: 50 Time: 2.5 hours

[Answer all questions from **Group-A** and **Group-B**
Separate answer script must be used for Group-A and Group-B.]

Group-A

1.a)	Discuss the clocking methodology for datapath implementation.	3	CO1
b)	Suppose you are given an instruction **sub $s2,$t2,$t3**. Describe the complete operation of the datapath using the instruction with a figure.	4	CO1
	Or,		
	Suppose you are given an instruction **lw $s1,100($s3)**. Describe the complete operation of the datapath using the instruction with a figure		
c)	What are the advantages of multicycle implementation over single cycle?		
2.a)	Design a single cycle data path for handling R-type instruction of MIPS.	3	CO
	Or,	4	CO
	Design a multi-cycle data path for handling basic instruction of MIPS.		
b)	Describe all temporary register and multiplexors of Multicycle Datapath.	4	CO
c)	Do you think that the functions of memory data register and instruction register are the same? If not then how?	2	CO
	Or,		
	Why a single-cycle implementation is not used? justify the answer.		

Group-B

3.a)	Does any Pipeline Stall arise here for the following instructions----- explain: **MUL R1, R2, R3** **SUB R3, R1, R4** **ADD R4,R5,R6**	3	CO
b)	What do you mean by pipelining? Describe advantages of pipelining architecture.	5	CO
c)	Dynamic pipelining is more complecated than the traditional or static pipelining – why?	2	CO
4.a)	Write short note on structural hazard. Or, Write short note on memory hierarchy.	2	CO
b)	Explain interrupt and its classes with necessary diagram. Or, What is TLB? Discuss the operation of paging and TLB using flow diagram?	4	CO
c)	What is DMA controller? Describe the configuration of DMA controller.		
5.a)	Discuss the clocking methodology with relevant diagram	4	CO
b)	Explain the handshaking protocol with diagram. Or, Explain the direct mapped cache system with figure.	6 4	CO CO

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Total marks: 50 Time: 2.5 hours

[Answer all the questions; Separate answer script must be used for **Group-A** and **Group-B**.]

Group-A

1.a)	What are the basic elements in an MIPS data path implementation?	3	CO1
b)	Suppose you are given an instruction sub $s2,$t2,$t3. Describe the complete operation of the datapath using the instruction with a figure.	4	CO1
c)	Why a single-cycle implementation is not used? justify the answer	3	CO2
2.a)	Design a single cycle data path for handling R-type instruction of MIPS.	3	CO1
b)	Describe all temporary register and multiplexors of Multicycle Datapath	4	CO2
c)	Do you think that the functions of memory data register and instruction register are the same? If not then how?	3	CO3
	Or,		
2. a)	Are instruction and microinstruction same? If not then why?	3	CO1
b)	Illustrate the following instruction with three bus and two bus CPU system and explain which is better? ADD R3, R2, R1	4	CO2
c)	Set the control words for the following microinstructions: i) SUB R1, R2, R0, ii) MAR← PC	3	CO3

Group-B

3.a)	Does any Pipeline Stall arise here for the following instructions, explain: MUL R1, R2, R3 SUB R3, 1, R4 ADD R4, R5, R6	4	CO2
b)	Which is advantageous between sequential processing and pipeline processing, explain?	2	CO2
c)	How does data dependency effect in pipelining?	4	CO2
4.a)	Suppose you have 6 I/O devices in your computer. An output operation will be performed to the Device 4 (could be the keyboard) in the case of shared I/O arrangement. Explain with diagram how the operation will be performed.	5	CO3
b)	Suppose you would like to bi-pass the CPU to perform a job. How a DMA can work in this regard?	5	CO1
5.a)	What is DMA controller? Describe the working principle of DMA controller.	3	CO1
b)	What is TLB? How can you reduce page fault explain logically?	3	CO3
c)	Write short note on handshaking protocol.	4	CO1
	Or,		
5a)	Write short note on memory hierarchy.	3	CO1
b)	Define *cache hit, cache miss, hit rate, tag* in caching.	4	CO1
c)	Suppose you would like to bi-pass the CPU to perform a job. How a DMA can work in this regard ?	3	CO3

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